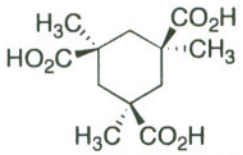
Organic Cume
2-11-11

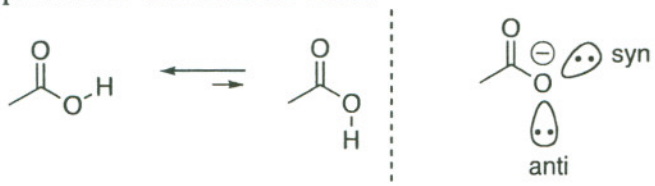
1. Sketch the π orbitals for allyl anion and for an enolate anion, and discuss the differences.

2. Predict the most stable conformation of the following molecule, called Kemp's triacid.



3. In the following heterocyclic compounds, the keto form dominates over the enol form in solution. Suggest a reason for this.

4. The preferred conformation of carboxylic acids is called s-*trans*, where the hydrogen on the OH is trans to the R group, as shown for acetic acid below. *Ab initio* computational modeling finds a 5.9 kcal/mol preference in the gas phase for acetic acid. However, further computational work modeling aqueous solvation found only a 1.7 kcal/mol preference. First, state which lone pair in a carboxylate anion is the more basic, the syn or anti lone pair. Second, explain why there is a difference in the equilibria between the gas phase and water, and why the preference is smaller in water.



5. The **Thorp-Ingold effect** (also called the **gem-dimethyl effect**) is an effect on reactivity in cyclizations due to geminal methyl groups. Draw the differing conformations of 3,3-dimethylpentane, and predict the lowest energy conformation. Use the terms syn, anti, clinal, and periplanar to define the relationships between the main chain carbons. Make a guess as to what the Thorp-Ingold effect is.